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Isolated Versus Non-Isolated Fixed Point, 2D Dynamics

I am trying to understand the classification of fixed points in a dynamical systems context (fixed points of a system of two linear differential equations are places where both $x'_1 = x'_2 = 0$).

Representing 2D systems as a matrix equation $\vec{x}' = A \vec{x}$, Strogatz classifies fixed points based on τ , the trace of the matrix and Δ , the determinant of the matrix:

- unstable node: $\tau > \sqrt{4\Delta}$, $\Delta > 0$
- unstable spiral (spiral source): $\sqrt{4\Delta} > \tau > 0$, $\Delta > 0$
- neutrally stable centers: $\tau = 0$, $\Delta > 0$
- stable spiral (spiral sink): $-\sqrt{4\Delta} < \tau < 0$, $\Delta > 0$
- stable node: $\tau < -\sqrt{4\Delta}$, $\Delta > 0$
- saddle point: $\Delta < 0$

However, he is very vague about the boundary cases. Specifically, what happens on the parabola $\tau^2 - 4\Delta = 0$ and the line $\Delta = 0$?

Strogatz mentions that these include star nodes (decoupled systems), degenerate nodes (one unique eigendirection), and nonisolating fixed points. However, in a later problem, he mentions "isolating fixed points". What is the difference?

How are all of these nonstandard edge cases classified in terms of τ and Δ ?

(differential-equations) (dynamical-systems) (chaotic-systems)

edited May 17 '13 at 6:11

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asked Mar 22 '13 at 1:17

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1 Answer

An isolated fixed point means that one can construct a region around the fixed point such that no other fixed points lie within. A nonisolated fixed point is the converse (i.e. there are other fixed points arbitrarily close; in practice, these end up being lines or a plane of fixed points).

As far as classification:

For 2D linear systems (or for linearization *predictions* concerning 2D nonlinear systems):

- if $\Delta < 0$:
Isolated fixed point
CASE #1: Saddle Point
- if $\Delta = 0$:
Nonisolated fixed points
 - if $\tau < 0$:
CASE #2: Line of Ляпунов (Lyapunov) stable fixed points
 - if $\tau = 0$:
CASE #3: Plane of fixed points
 - if $\tau > 0$:
CASE #4: Line of unstable fixed points
- if $\Delta > 0$:
Isolated fixed point
 - if $\tau < -\sqrt{4\Delta}$:
CASE #5: Stable Node
 - if $\tau = -\sqrt{4\Delta}$:
 - if there are no uniquely determined eigenvectors (both can be anything):
CASE #6: Stable Star
 - if there is one uniquely determined eigenvector (the other can be anything):
CASE #7: Stable Degenerate Node

- if $-\sqrt{4\Delta} < \tau < 0$:
CASE #8: Stable Spiral
- if $\tau = 0$:
CASE #9: Stable Center
- if $0 < \tau < \sqrt{4\Delta}$:
CASE #10: Unstable Spiral
- if $\tau = \sqrt{4\Delta}$:
 - if there are no uniquely determined eigenvectors (both can be anything):
CASE #11: Unstable Star
 - if there is one uniquely determined eigenvector (the other can be anything):
CASE #12: Unstable Degenerate Node
- if $\sqrt{4\Delta} < \tau$:
CASE #13: Unstable Node

General Notes:

- For 2D linear systems, the above predictions are always accurate.
- For 2D nonlinear systems, when the above are used as predictions:
 - The descriptions are always correct for cases #1, #5, #8, #10, and #13 but can be inaccurate for cases #2, #3, #4, #6, #7, #9, #11, and #12.
 - Ambiguous cases #6, #7, #11, and #12 at least have their stability correctly determined.
 - If the system is conservative, a prediction of case #9 is accurate.

answered Apr 28 '13 at 22:18



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